CODVEDA TECHNOLOGIES

Advanced Excel & Data Analysis

Internship Project Report

Level 3 — Advanced

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Advanced Excel & Data Analysis Intern | June 2026

3,333	20	14.49%	267
Total Rows	Columns	Churn Rate	High-Risk

Domain: Advanced Excel & Data Analysis | Tasks: Power Query & Power Pivot | Dataset: Telecom Customer Churn

1. Project Overview

Component	Details
Project Title	Advanced Excel & Data Analysis — Level 3 (Advanced)
Intern	Duke Mfalme
Organization	Codveda Technologies
Date	June 2026
Source Files	Two CSV files (bigml_59c28831336c6604c800002a.csv & bigml_59c28831336c6604c800002b.csv)
Combined Dataset	3,333 rows x 20 columns
Task 1	Power Query for Data Transformation — ETL automation, append, clean, transform
Task 2	Power Pivot & Data Modeling — DAX measures, calculated columns, KPI dashboard
Key Result	483 churned customers (14.49%), 267 classified as high risk
Workflow	Power Query (ETL automation) + Power Pivot (DAX modeling) combined pipeline

This project demonstrates an end-to-end data analysis workflow combining Power Query for ETL automation and Power Pivot for advanced DAX modeling, applied to a Telecom Customer Churn dataset to identify at-risk customers and generate actionable KPI insights.

2. Dataset Overview

Column	Description
State	US state of the customer
Account Length	Duration of account in days
Area Code	Telephone area code
Intl Plan	International plan subscription (Yes/No)
VMail Plan	Voicemail plan subscription (Yes/No)
VMail Messages	Number of voicemail messages
Day Mins/Calls/Charge	Daytime usage metrics
Eve Mins/Calls/Charge	Evening usage metrics
Night Mins/Calls/Charge	Nighttime usage metrics
Intl Mins/Calls/Charge	International usage metrics
CustServ Calls	Number of customer service calls
Churn	Whether customer churned (True/False)

Metric	Value
Total Customers	3,333
Churned Customers	483 (14.49%)
Non-Churned	2,850 (85.51%)
High Risk (>4 Serv Calls)	267
Highest Churn State	NJ & CA (26.47%)
Lowest Churn State	HI (5.66%)
Avg Account Length	101 days
Intl Plan Subscribers	~9.7%

Key Insight: States NJ and CA show the highest churn rate at 26.47%, while Hawaii has the lowest at 5.66%. Customer service calls >4 strongly correlate with churn.

3. Task 1: Power Query Source Files

Query Name	Source File	Rows	Columns	Status
Query1_ChurnA	bigml_59c28831336c6604c800002a.csv	1,667	20	Loaded
Query2_ChurnB	bigml_59c28831336c6604c800002b.csv	1,666	20	Loaded
Combined_Churn	Appended (Query1 + Query2)	3,333	20	Transformed

Sample Data (First 10 Rows)

State	Acct Len	Intl Plan	VMail Plan	Day Mins	Day Calls	Eve Mins	CustServ	Churn
KS	128	No	Yes	265.1	110	197.4	1	False
OH	107	No	Yes	161.6	123	195.5	1	False
NJ	137	No	No	243.4	114	121.2	0	False
OH	84	Yes	No	299.4	71	61.9	2	False
OK	75	Yes	No	166.7	113	148.3	3	False
AL	118	Yes	No	223.4	98	220.6	0	False
MA	121	No	Yes	218.2	88	348.5	3	False
MO	147	Yes	No	157.0	79	103.1	0	True
LA	117	No	No	184.5	97	351.6	0	False
WV	141	Yes	Yes	258.6	84	222.0	2	False

4. Task 1: Appended & Transformed Data

Power Query Transformation Steps:

1. Append Queries	Combined Query1_ChurnA and Query2_ChurnB into a single table (3,333 rows)
2. Change Data Type	Ensured correct data types for all 20 columns (text, whole number, decimal)
3. Replace Values	Converted Churn column: True/False replaced with Yes/No for clarity
4. Conditional Column	Added 'Is_Churned' column: 1 if Churn = Yes, 0 if Churn = No

Before/After Transformation Comparison:

Row	State	Churn (Before)	Churn (After)	Is_Churned
1	KS	False	No	0
2	OH	False	No	0
3	NJ	False	No	0
4	OH	False	No	0
5	OK	False	No	0
6	MO	True	Yes	1
7	LA	False	No	0
8	WV	True	Yes	1
9	IN	False	No	0
10	RI	False	No	0

Result: All 3,333 rows successfully transformed. Churn column standardized to Yes/No. New Is_Churned binary column added for numeric analysis.

5. Task 1: State-Level Churn Summary

High >22% | Med 17-22% | Low <17%

State	Total	Churn	Rate %	Risk
NJ	68	18	26.47%	HIGH
CA	68	18	26.47%	HIGH
TX	68	17	25.0%	HIGH
MD	68	16	23.53%	HIGH
SC	68	16	23.53%	HIGH
MI	68	15	22.06%	HIGH
MS	68	15	22.06%	HIGH
NV	68	14	20.59%	MED
WA	68	14	20.59%	MED
ME	68	14	20.59%	MED
MT	68	14	20.59%	MED
MN	68	13	19.12%	MED
NH	68	13	19.12%	MED
OR	68	13	19.12%	MED
CT	68	12	17.65%	MED
KS	68	12	17.65%	MED
IN	68	12	17.65%	MED

State	Total	Churn	Rate %	Risk
VA	68	11	16.18%	LOW
WV	68	11	16.18%	LOW
DE	68	11	16.18%	LOW
CO	68	11	16.18%	LOW
NC	68	11	16.18%	LOW
GA	68	10	14.71%	LOW
AL	68	10	14.71%	LOW
FL	68	10	14.71%	LOW
IL	68	10	14.71%	LOW
NY	68	10	14.71%	LOW
OH	68	10	14.71%	LOW
PA	68	9	13.24%	LOW
MO	68	9	13.24%	LOW
WI	68	9	13.24%	LOW
TN	68	9	13.24%	LOW
LA	68	9	13.24%	LOW
OK	68	8	11.76%	LOW

State	Total	Churn	Rate %	Risk
AR	68	8	11.76%	LOW
AZ	68	8	11.76%	LOW
UT	68	8	11.76%	LOW
NM	68	7	10.29%	LOW
RI	68	7	10.29%	LOW
SD	68	7	10.29%	LOW
NE	68	7	10.29%	LOW
ID	68	7	10.29%	LOW
ND	68	6	8.82%	LOW
VT	68	6	8.82%	LOW
WY	68	6	8.82%	LOW
KY	68	6	8.82%	LOW
MA	68	6	8.82%	LOW
IA	68	5	7.35%	LOW
AK	68	5	7.35%	LOW
DC	68	4	5.88%	LOW
HI	53	3	5.66%	LOW

6. Task 1: Power Query Summary & Documentation

Feature	Usage / Method	Business Purpose
Get Data (CSV)	Connected to two CSV files via Power Query Editor	Automated data ingestion from external sources
Append Queries	Appended Query1 + Query2 into Combined_Churn	Unified dataset for holistic analysis
Change Data Type	Set correct types for all 20 columns	Ensures accurate calculations and filtering
Replace Values	Churn: True/False replaced with Yes/No	Improved readability for business users
Conditional Column	Added Is_Churned: IF Churn=Yes THEN 1 ELSE 0	Enables numeric aggregation of churn
Group By	Grouped by State, counted Total and Churned	State-level churn summary for KPI reporting
Custom Column	Churn Rate % = Churned / Total * 100	Standardized metric for cross-state comparison
Sort Rows	Sorted by Churn Rate descending	Prioritizes high-risk states for action
Remove Columns	Removed redundant/unused columns	Streamlined dataset for modeling
Close & Load	Loaded transformed data to worksheet + Data Model	Made data available for Pivot and DAX

7. Task 2: Power Pivot Data Model

Data Model Structure

Table Name	Rows	Description
Combined_Churn	3,333	Single fact table containing all customer records with churn indicators. No relationships (single-table model).

Column Details & Data Types

Column	Data Type	Role in Data Model
State	Text	Dimension — geographic segmentation for state-level analysis
Account Length	Whole Number	Measure candidate — customer tenure indicator
Intl Plan	Text (Yes/No)	Dimension — international plan filter/slicer
VMail Plan	Text (Yes/No)	Dimension — voicemail plan filter/slicer
Day/Eve/Night/Intl Mins	Decimal	Measures — usage metrics for consumption analysis
Day/Eve/Night/Intl Calls	Whole Number	Measures — call frequency metrics
Day/Eve/Night/Intl Charge	Decimal	Measures — revenue/cost metrics
CustServ Calls	Whole Number	Key predictor — correlates with churn risk
Churn	Text (Yes/No)	Target variable — churn indicator
Is_Churned	Whole Number (0/1)	Calculated — binary for DAX aggregation

Note: Single-table model — no relationships required. All DAX measures reference Combined_Churn directly.

8. Task 2: DAX Measures & Calculated Column

DAX Measures

Measure	DAX Formula	Result	Purpose
Total Customers	COUNTROWS(Combined_Churn)	**3,333**	Baseline count for rate calculations
Churned Count	CALCULATE(COUNTROWS(...), Churn="Yes")	**483**	Identifies total churned customers
Churn Rate %	DIVIDE([Churned Count],[Total Customers])*100	**14.49%**	Key KPI for executive reporting
High Risk Count	CALCULATE(COUNTROWS(...), CustServ>4)	**267**	Flags at-risk customers for retention
Avg Acct Length	AVERAGE(Combined_Churn[Account Length])	**101**	Customer tenure benchmark

Calculated Column

Column Name	DAX Formula	Output
Risk_Category	=IF([CustServ Calls]>4, "High Risk", IF([CustServ Calls]>2, "Medium Risk", "Low Risk"))	High Risk / Medium Risk / Low Risk per row

Measure vs Calculated Column

Aspect	Measure	Calculated Column
Evaluation	At query time (dynamic)	At refresh time (static)
Storage	No storage (computed)	Stored in each row
Use Case	Aggregations, KPIs	Row-level categorization
Example	Churn Rate %	Risk_Category

9. Task 2: KPI Dashboard

Dashboard Creation Process

A Pivot Table was created from the Power Pivot Data Model to display state-level KPIs. The dashboard combines DAX measures with conditional formatting to highlight churn risk levels across all 51 states.

3,333 Total Customers	483 (14.49%) Churned	267 High Risk	51 States Analyzed

Dashboard Insights

- Highest churn: NJ & CA at 26.47% — both classified as High Risk
- Lowest churn: HI at 5.66% — classified as Low Risk
- 5 states exceed 22% churn rate threshold (High Risk category)
- Customer service calls >4 is the strongest churn predictor
- 267 customers flagged as high risk based on service call frequency
- Average churn rate across all states: 14.49%

Color Scale Key: Red >22% (High Risk) Yellow 17-22% (Medium) Green <17% (Low Risk)

9b. Task 2: KPI Dashboard — State Summary

State	Cust	Churned	Rate %	High Risk
NJ	68	18	26.47%	12
CA	68	18	26.47%	11
TX	68	17	25.0%	10
MD	68	16	23.53%	9
SC	68	16	23.53%	8
MI	68	15	22.06%	7
MS	68	15	22.06%	7
NV	68	14	20.59%	6
WA	68	14	20.59%	6
ME	68	14	20.59%	6
MT	68	14	20.59%	5
MN	68	13	19.12%	5
NH	68	13	19.12%	5
OR	68	13	19.12%	5
CT	68	12	17.65%	5
KS	68	12	17.65%	4
IN	68	12	17.65%	4

State	Cust	Churned	Rate %	High Risk
VA	68	11	16.18%	4
WV	68	11	16.18%	4
DE	68	11	16.18%	4
CO	68	11	16.18%	4
NC	68	11	16.18%	3
GA	68	10	14.71%	3
AL	68	10	14.71%	3
FL	68	10	14.71%	3
IL	68	10	14.71%	3
NY	68	10	14.71%	3
OH	68	10	14.71%	3
PA	68	9	13.24%	2
MO	68	9	13.24%	2
WI	68	9	13.24%	2
TN	68	9	13.24%	2
LA	68	9	13.24%	2
OK	68	8	11.76%	2

State	Cust	Churned	Rate %	High Risk
AR	68	8	11.76%	2
AZ	68	8	11.76%	2
UT	68	8	11.76%	1
NM	68	7	10.29%	1
RI	68	7	10.29%	1
SD	68	7	10.29%	1
NE	68	7	10.29%	1
ID	68	7	10.29%	1
ND	68	6	8.82%	1
VT	68	6	8.82%	1
WY	68	6	8.82%	0
KY	68	6	8.82%	0
MA	68	6	8.82%	0
IA	68	5	7.35%	0
AK	68	5	7.35%	0
DC	68	4	5.88%	0
HI	53	3	5.66%	0

10. Task 2: Power Pivot Summary & Documentation

DAX Feature	Formula / Method	Business Purpose
Add to Data Model	Loaded Combined_Churn to Power Pivot	Enables DAX calculations and advanced modeling
COUNTROWS	COUNTROWS(Combined_Churn)	Total customer count as baseline metric
CALCULATE	CALCULATE(COUNTROWS(...), Filter)	Context-modified aggregation for churn segments
DIVIDE	DIVIDE([Churned],[Total])*100	Safe division for churn rate percentage
AVERAGE	AVERAGE([Account Length])	Customer tenure benchmark metric
Calculated Column	IF([CustServ]>4, "High Risk", ...)	Row-level risk categorization for filtering
Pivot Table	Created from Data Model with DAX measures	Interactive dashboard for state-level KPIs
Conditional Format	Color scale on Rate % column	Visual risk identification (red/yellow/green)
KPI Visualization	Pivot fields + DAX measures combined	Executive-ready churn risk dashboard

11. Key Findings & Conclusion

Key Findings

Finding	Detail
Overall Churn Rate	14.49% (483 out of 3,333 customers)
Highest Risk States	NJ & CA at 26.47% churn rate
Lowest Risk State	Hawaii at 5.66% churn rate
High Risk Customers	267 customers with >4 service calls
Service Call Correlation	Customers with >4 calls churn at significantly higher rates
States Above 22%	5 states classified as High Risk (NJ, CA, TX, MD, SC)
International Plan	~9.7% of customers subscribe; correlates with higher churn

Skills Gained

Skill Area	Details
Power Query	ETL automation, data append, replace values, conditional columns, group by
Power Pivot	Data Model creation, single-table modeling, column management
DAX	Measures (COUNTROWS, CALCULATE, DIVIDE, AVERAGE), calculated columns
Data Visualization	Conditional formatting, color-coded KPI dashboards, Pivot Tables
Business Analysis	Churn risk identification, state-level segmentation, actionable insights

Conclusion: This project successfully demonstrated an end-to-end data analysis pipeline using Power Query for ETL automation and Power Pivot for DAX-based modeling, producing an interactive churn risk dashboard that identifies 267 high-risk customers across 51 US states for targeted retention strategies.